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1. Introduction

As a project, I decided to design and implement a simple Tank Battle Game. The idea of this game is based on classic 2-D tank games where the player controls a tank, moves around the map, and destroys enemy tanks. This project is suitable for object-oriented programming because the game has many objects such as tanks, bullets, walls, enemies, and the game map.

The main goal of this project is to create a small but playable game using Java. The game is not too complicated, but it still includes important programming concepts such as classes, inheritance, arrays/lists, keyboard input, collision detection, and a game loop.

2. Overview

2.1 Gameplay

The player controls a green tank. The player can move up, down, left, and right by using the keyboard. The player can also shoot bullets to destroy red enemy tanks. Enemy tanks move by themselves and sometimes shoot at the player. The player wins when all enemy tanks are destroyed.

2.2 Map

The game map is a 2-D area with walls placed in different positions. These walls act as obstacles. Tanks cannot move through walls, and bullets are also stopped when they hit a wall.

2.3 Player Initial Items

- The player starts with 100 health points.
- The player has unlimited bullets.
- The player starts near the bottom of the screen.
- The score starts at 0.

2.4 Modes

The first version of the game has a single-player mode. One player fights against multiple enemy tanks. Multiplayer mode can be added later as a future improvement.

2.5 Bullets

Bullets move in the direction where the tank is facing. If a bullet hits an enemy, the enemy loses health. If the enemy health becomes zero, the enemy is removed from the game and the score increases.

2.6 Tanks

There are two types of tanks: the player tank and enemy tanks. The player tank is controlled by keyboard input. Enemy tanks use simple random movement and random shooting.

3. Functional Requirements

ID	Requirement
FR1	The player shall be able to move the tank using W, A, S, and D keys.
FR2	The player shall be able to shoot using the space bar.
FR3	The system shall display health, score, and enemies

	left.
FR4	Enemy tanks shall move automatically.
FR5	Enemy tanks shall shoot bullets at random times.
FR6	Walls shall block tanks and bullets.
FR7	The game shall end when the player health reaches zero.
FR8	The game shall show a win screen when all enemies are destroyed.
FR9	The player shall be able to restart the game by pressing R.

4. Non-Functional Requirements

- Usability: Controls are simple and easy to learn.
- Performance: The game updates frequently using a timer for smooth movement.
- Maintainability: Classes are separated based on their responsibilities.
- Portability: The program can run on any computer with Java installed.
- Understandability: The code uses clear class names such as Tank, Bullet, Wall, and EnemyTank.

5. System Models

5.1 Use Case Model

Use Case	Description
Play Game	Player starts the game and controls the tank.
Move Tank	Player moves the tank using keyboard keys.
Shoot Bullet	Player shoots to destroy enemies.
Enemy Attack	Enemy tanks move and shoot automatically.
Restart Game	Player presses R to restart after winning or losing.

5.2 Basic Event Flow: Play Game

- Player opens the game window.
- System creates player tank, enemy tanks, walls, and bullets list.
- Player moves and shoots using keyboard controls.
- System updates tank movement, enemy movement, bullets, and collisions.
- If all enemies are destroyed, the system displays You Win.
- If player health reaches zero, the system displays Game Over.

5.3 Object/Class Model

Class	Purpose
TankBattleGame	Main class that controls the window, timer, movement, score, and collisions.
Tank	Parent class for tanks. It stores position, size, speed, health, direction, and drawing logic.
PlayerTank	Child class of Tank used for the user-controlled tank.
EnemyTank	Child class of Tank used for enemy movement and shooting.
Bullet	Represents bullets fired by player or enemies.

Wall	Represents obstacles on the game map.
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6. Conclusion

The Tank Battle Game is a small but complete object-oriented game project. It includes the main features needed for a playable game: player control, enemy tanks, shooting, collisions, health, score, winning, losing, and restarting. The project also leaves room for future improvements such as multiplayer mode, more levels, sound, better graphics, and bonus items.

7. Improvement Summary

- Add more levels with different maps.
- Add sound effects for shooting and explosions.
- Add bonus items such as health and speed.
- Add multiplayer mode.
- Improve enemy AI.

8. Glossary

Term	Meaning
JPanel	Java Swing component used for drawing the game.
Timer	Java Swing timer used for the game loop.
Collision Detection	Checking when objects touch each other.
Game Loop	Repeated update process that moves objects and redraws the screen.